Thermal Finite Element Analysis of a Slotted Electric Motor Cross-Section

Physics, Mathematics, Numerical Method, and Results

Project Technical Report

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Abstract

This report documents a 2D finite element model for steady-state thermal analysis of an electric motor cross-section. The objective is to solve heat conduction with internal winding losses and convective cooling, while preserving clear physical interpretation and numerical robustness. The model includes: (i) motor-region material partitioning (iron, slot/winding effective medium, air gap), (ii) optional temperature-dependent winding loss, (iii) Robin convective boundaries on both outer stator and inner air-gap sides, and (iv) validation against an analytical annulus benchmark. We summarize the governing physics, derive the weak form, describe the finite element algorithm, and present key simulation results and sensitivity trends.

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1 Problem Overview and Scope

The project solves two thermal problems:

1. **Phase 1 (verification problem):** A plain annulus with prescribed inner/outer temperatures, compared to analytical solution.
2. **Phase 2 (application problem):** A slotted motor cross-section with volumetric winding heating and convective cooling.

The implementation is intentionally transparent: plain Python functions, sparse matrix assembly, and direct sparse solve. The report focus is on physics and mathematics first, then implementation and results.

Modeling assumptions

- Steady state (no transient terms).
- Two-dimensional cross-section; results are per unit axial length (W/m).
- No radiation term in the base formulation.
- Effective material properties are used for slot region and laminated iron approximation.

2 Motor Thermal Physics

2.1 Governing equation

In each material region Ω_m, the steady heat equation is

$$-\nabla \cdot (\mathbf{K}(\mathbf{x}, T)\nabla T) = q(\mathbf{x}, T), \tag{1}$$

where T is temperature, $\mathbf{K}$ is thermal conductivity (scalar or tensor), and q is volumetric heat source.

For isotropic regions, $\mathbf{K} = k\mathbf{I}$. For effective anisotropic iron in polar orientation, a 2×2 tensor is used in Cartesian coordinates.

2.2 Boundary physics

Two Robin boundaries are used in Phase 2:

$$-\mathbf{n} \cdot (\mathbf{K}\nabla T) = h\,(T - T_\infty)\,, \tag{2}$$

with separate (h, T_∞) values for:

- Outer stator boundary (external cooling).
- Inner air-gap boundary (effective rotor-side cooling).

2.3 Temperature-dependent winding loss

For winding elements, Joule heating can be updated with temperature:

$$q_{\mathrm{cu}}(T) = q_{\mathrm{ref}}\left[1 + \alpha_{\mathrm{cu}}(T - T_{\mathrm{ref}})\right], \tag{3}$$

with practical clipping bounds for numerical stability in prototype studies.

3 Schematics of the Motor Problem

3.1 Domain and region schematic

Figure 1 shows a conceptual cross-section and region partition.

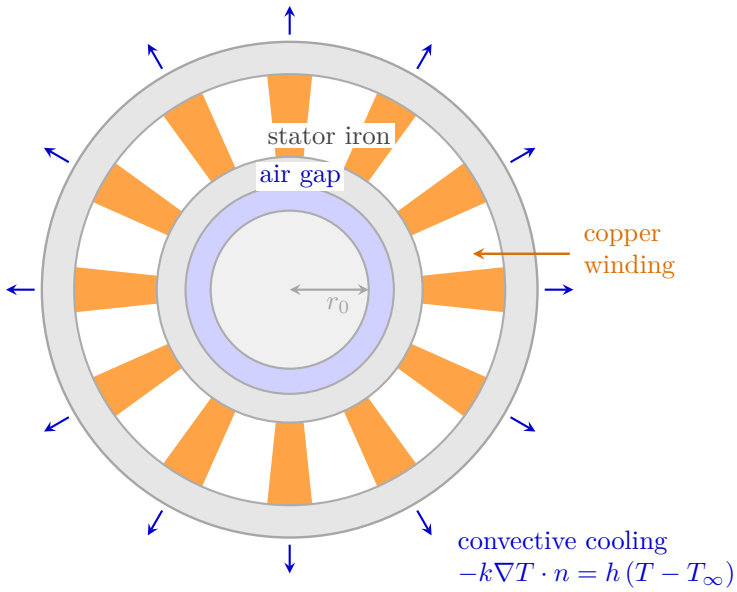


Figure 1: Schematic of the modeled motor cross-section and thermal boundary conditions.

3.2 Heat-flow schematic

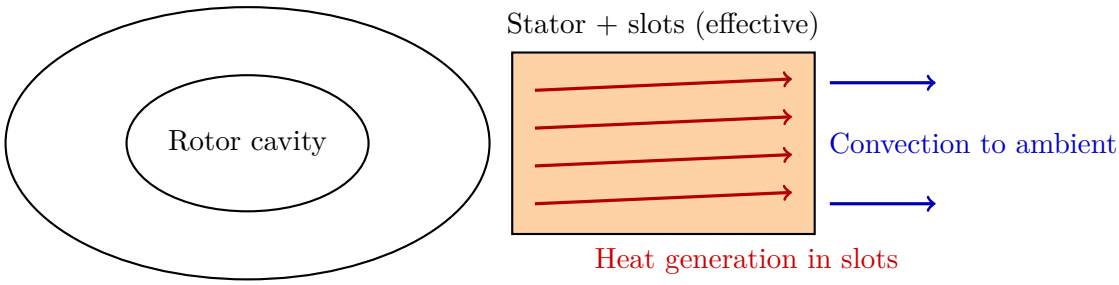


Figure 2: Conceptual thermal pathway: winding losses conduct through solids and are removed by convection.

4 Finite Element Formulation

4.1 Weak form

Let v be a test function. Integrating over Ω and applying integration by parts gives

$$\int_\Omega (\nabla v)^T \mathbf{K} \nabla T \, d\Omega + \int_{\Gamma_R} hvT \, d\Gamma = \int_\Omega vq \, d\Omega + \int_{\Gamma_R} hvT_\infty \, d\Gamma. \tag{4}$$

This leads to linear algebra system

$$\mathbf{K}_g \mathbf{T} = \mathbf{f}, \tag{5}$$

where $\mathbf{K}_g$ is sparse and assembled from elemental contributions.

4.2 Element equations (P1 triangles)

For linear triangular elements, shape-function gradients are constant inside each element:

$$\nabla N_i = [b_i, \ c_i]^T. \tag{6}$$

Element stiffness entries are

$$K_e(i,j) = A_e \, \nabla N_i^T \mathbf{K}_e \nabla N_j. \tag{7}$$

Element load entries for uniform source are

$$f_e(i) = q_e A_e / 3. \tag{8}$$

4.3 Robin boundary edge contribution

For a boundary edge of length L:

$$\mathbf{K}_{R,e} = \frac{hL}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{f}_{R,e} = \frac{hT_\infty L}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \tag{9}$$

4.4 Sparse assembly and solve

The implementation uses COO triplet accumulation and conversion to CSR format, then direct sparse solve (SuperLU). This is robust for the problem sizes used in this project.

5 Mesh Strategy and Region Tagging

5.1 Phase 1 mesh

Phase 1 uses a structured annular O-grid with two triangles per logical quadrilateral. This provides smooth refinement and clean convergence studies.

5.2 Phase 2 mesh

Phase 2 now uses an interface-conforming structured polar mesh with explicit angular breakpoints at slot boundaries and radial breakpoints at material interfaces. This minimizes interface smearing and avoids centroid-tag ambiguity common in unconstrained triangulation.

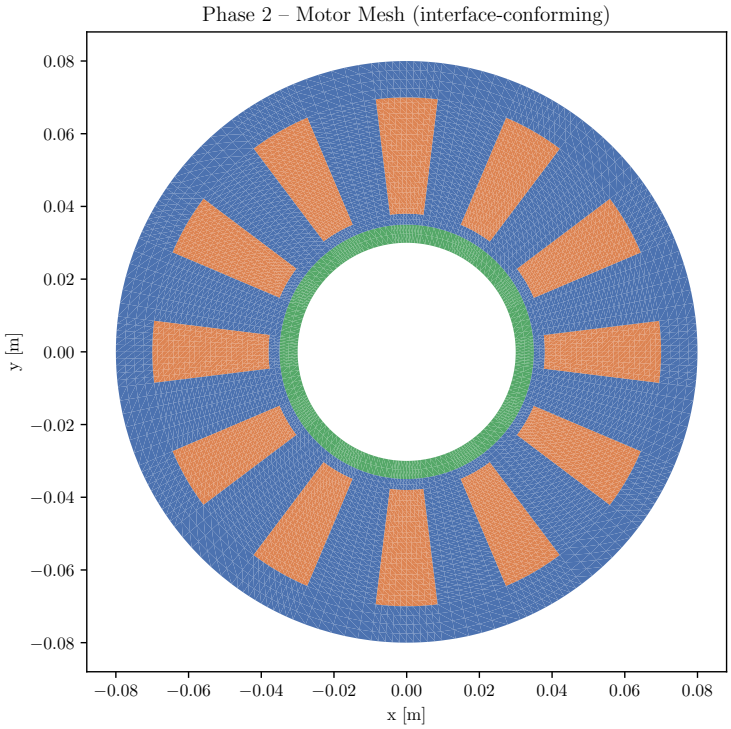


Figure 3: Generated motor mesh for Phase 2.

6 Algorithm

6.1 Overall workflow

1. Build mesh and region tags.
2. Build per-element conductivity model (tensor in iron, scalar elsewhere).
3. Initialize volumetric source q in slots.
4. Nonlinear fixed-point loop (if $q(T)$ enabled):
   (a) Assemble global system including Robin boundaries.
   (b) Solve for T.
   (c) Update slot $q(T)$, with under-relaxation and clipping.
   (d) Check convergence in ΔT and Δq.
5. Post-process: temperature maps, flux vectors, and energy balance checks.

6.2 Pseudo-code

Listing 1: Simplified Phase 2 solver logic

```
mesh = make_motor_mesh(...)
inner_edges = make_closed_boundary_edges(inner_nodes)
outer_edges = make_closed_boundary_edges(outer_nodes)

q = q_ref in slot elements
for k in range(max_iters):
    K, f = assemble(nodes, elements, conductivity_tensor, q)
    K, f = apply_robin(K, f, outer_edges, h_outer, Tinf_outer)
```

```
    K, f = apply_robin(K, f, inner_edges, h_gap, Tinf_gap)
    T = solve(K, f)

    q_new = update_q_from_temperature(T)
    q = (1-omega)*q + omega*q_new
    if converged(T, q):
        break

# final consistency solve with latest q
K, f = assemble(..., q)
T = solve(K, f)
```

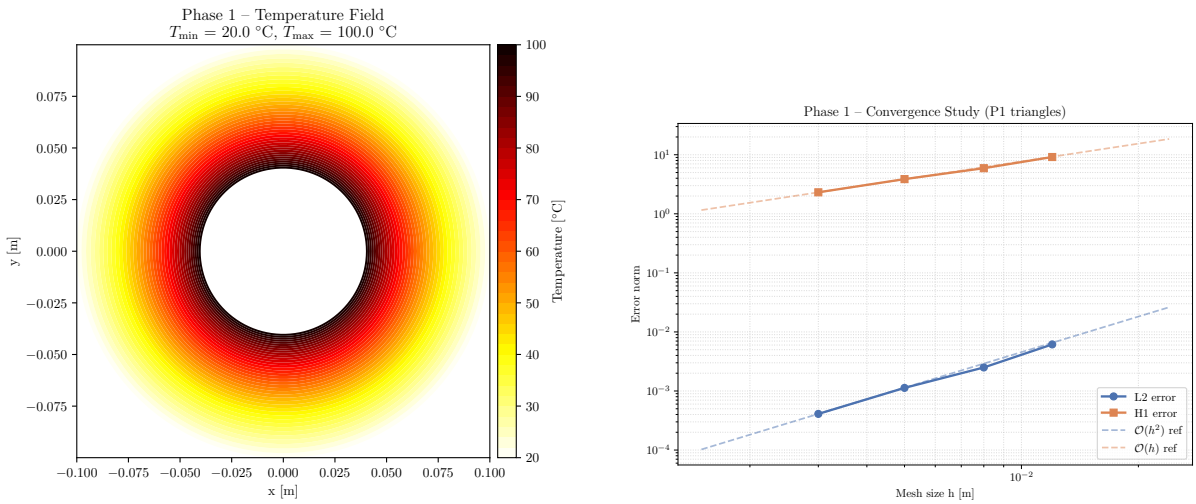
7 Results

7.1 Phase 1 verification (annulus)

The annulus benchmark validates implementation quality against analytical solution. Typical run outcomes:

- Max pointwise error: $6.67 \times 10^{-3}\,°\mathrm{C}$
- Measured convergence rates: $L_2 \approx 1.925$, $H^1 \approx 0.983$

These are close to theoretical rates for linear triangular elements ($O(h^2)$ in L_2, $O(h)$ in H^1).



(a) Phase 1 temperature field

(b) Phase 1 convergence

Figure 4: Verification plots from the annulus benchmark.

7.2 Phase 2 baseline (improved parameter set)

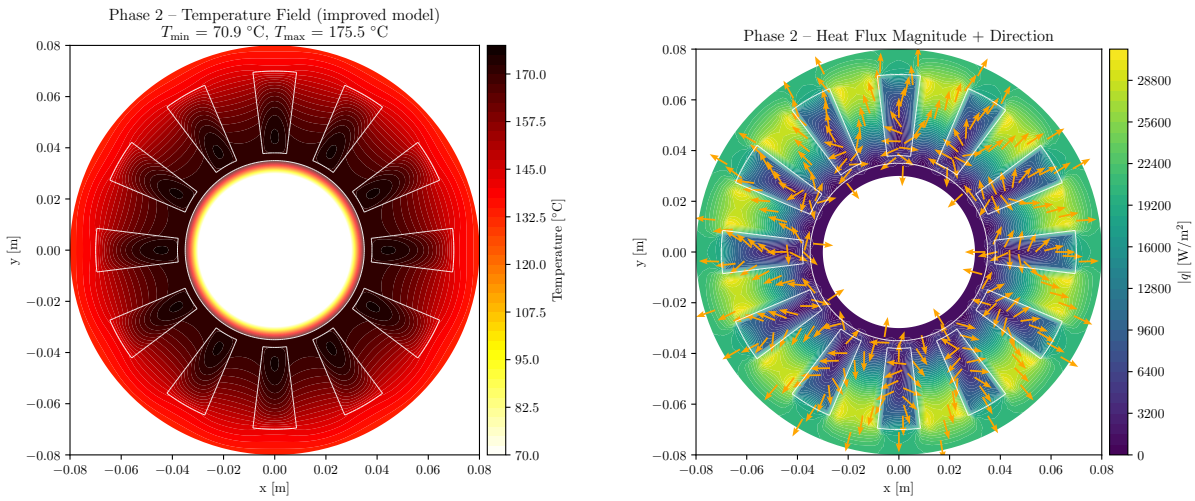
Using calibrated starter parameters $q_{ref} = 1.8 \times 10^6\,\mathrm{W/m^3}$, $h_{outer} = 180\,\mathrm{W/(m^2K)}$, $h_{gap} = 120\,\mathrm{W/(m^2K)}$, $k_{slot,eff} = 5\,\mathrm{W/(mK)}$, $k_r = 22$, $k_t = 30$, the model produced:

- Peak winding temperature $\approx 175.5\,°\mathrm{C}$
- Robin energy-balance residual $\approx 0.0\%$
- Independent conductive boundary check residual $\approx 1.4\%$

7.3 Sensitivity trends

The parametric trends are physically consistent:

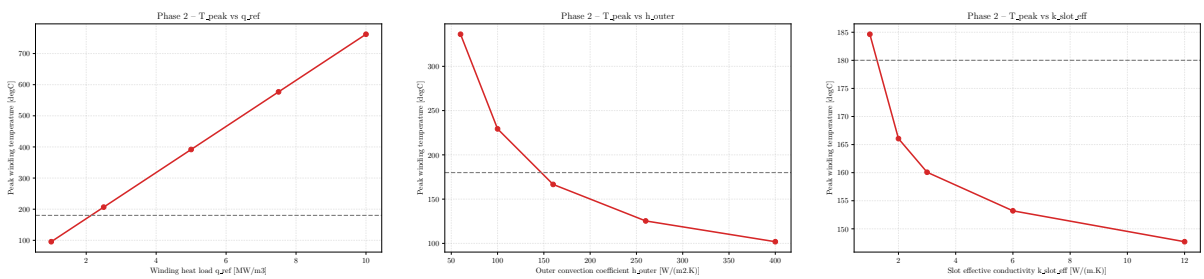
- T_{peak} increases with winding heat load.
- T_{peak} decreases with stronger outer convection.
- T_{peak} decreases with higher slot effective conductivity.



(a) Baseline temperature field

(b) Flux magnitude and direction

Figure 5: Phase 2 baseline fields.



(a) T_{peak} vs q_{ref}

(b) T_{peak} vs h_{outer}

(c) T_{peak} vs $k_{slot,eff}$

Figure 6: Phase 2 sensitivity studies.

8 Discussion

8.1 What is validated

- Mathematical correctness of FEM implementation is supported by Phase 1 convergence and analytical agreement.
- Energy bookkeeping is consistent in Phase 2 via two checks (Robin and independent conductive flux estimate).
- Updated mesh strategy improves interface fidelity and reduces artificial region mixing.

8.2 What remains approximate

- 2D model omits axial heat spreading and end-winding effects.
- Effective material properties and Robin coefficients need calibration to a specific motor hardware setup.
- Radiation and detailed CFD coupling are excluded in this prototype.

8.3 Recommended next steps

1. Calibrate h-values and effective conductivities with test data.
2. Add transient solve for duty-cycle analysis.
3. Introduce uncertainty bounds for key parameters and report confidence intervals for T_{peak}.

9 Conclusion

A compact but physically grounded thermal FEM workflow has been developed for a slotted motor cross-section. The model combines classical steady conduction theory, sparse P1 FEM discretization, practical nonlinear winding-loss updates, and robust post-processing diagnostics. Verification against analytical annulus theory shows expected convergence behavior. Application results provide credible design insight and sensitivity trends, while remaining suitable for iterative calibration and engineering decision support.